Angstrong Nuwa-HP60C

1. Content Description

> [!IMPORTANT]
>
> - This section content is only for users with the Angstrong Nuwa-HP60C camera package to reference

- This tutorial introduces the Angstrong Nuwa-HP60C used on the product, viewing color images, depth images, infrared images, and point clouds published by camera nodes.

2. Camera Introduction

3. ROS Drive Camera

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

Open a terminal inside the container,

```
exec_m3
```

- Run command in terminal

```
ros2 launch m3_bringup bringup_camera.launch.py
```

If the camera cannot be started, you need to check if the camera connection to the mainboard/Hub expansion board is loose.

You can use ros2 node tools to view which topics the camera node has published and which services are provided. Enter in terminal,

```
ros2 node info /ascamera_hp60c/camera_publisher
```

Published topics are as shown in the figure below

```
/ascamera_hp60c/camera_publisher
  Subscribers:
    /parameter_events: rcl_interfaces/msg/ParameterEvent
  Publishers:
    /ascamera_hp60c/camera_publisher/depth0/camera_info:
sensor_msgs/msg/CameraInfo
    /ascamera_hp60c/camera_publisher/depth0/image_raw: sensor_msgs/msg/Image
    /ascamera_hp60c/camera_publisher/depth0/points: sensor_msgs/msg/PointCloud2
    /ascamera_hp60c/camera_publisher/rgb0/camera_info:
sensor_msgs/msg/CameraInfo
    /ascamera_hp60c/camera_publisher/rgb0/image: sensor_msgs/msg/Image
    /parameter_events: rcl_interfaces/msg/ParameterEvent
    /rosout: rcl_interfaces/msg/Log
    /tf: tf2_msgs/msg/TFMessage
```

Provided services are as shown in the figure below,

```
  Service Servers:
    /ascamera_hp60c/camera_publisher/describe_parameters:
rcl_interfaces/srv/DescribeParameters
    /ascamera_hp60c/camera_publisher/get_parameter_types:
rcl_interfaces/srv/GetParameterTypes
    /ascamera_hp60c/camera_publisher/get_parameters:
rcl_interfaces/srv/GetParameters
    /ascamera_hp60c/camera_publisher/list_parameters:
rcl_interfaces/srv/ListParameters
    /ascamera_hp60c/camera_publisher/set_parameters:
rcl_interfaces/srv/SetParameters
    /ascamera_hp60c/camera_publisher/set_parameters_atomically:
rcl_interfaces/srv/SetParametersAtomically
```

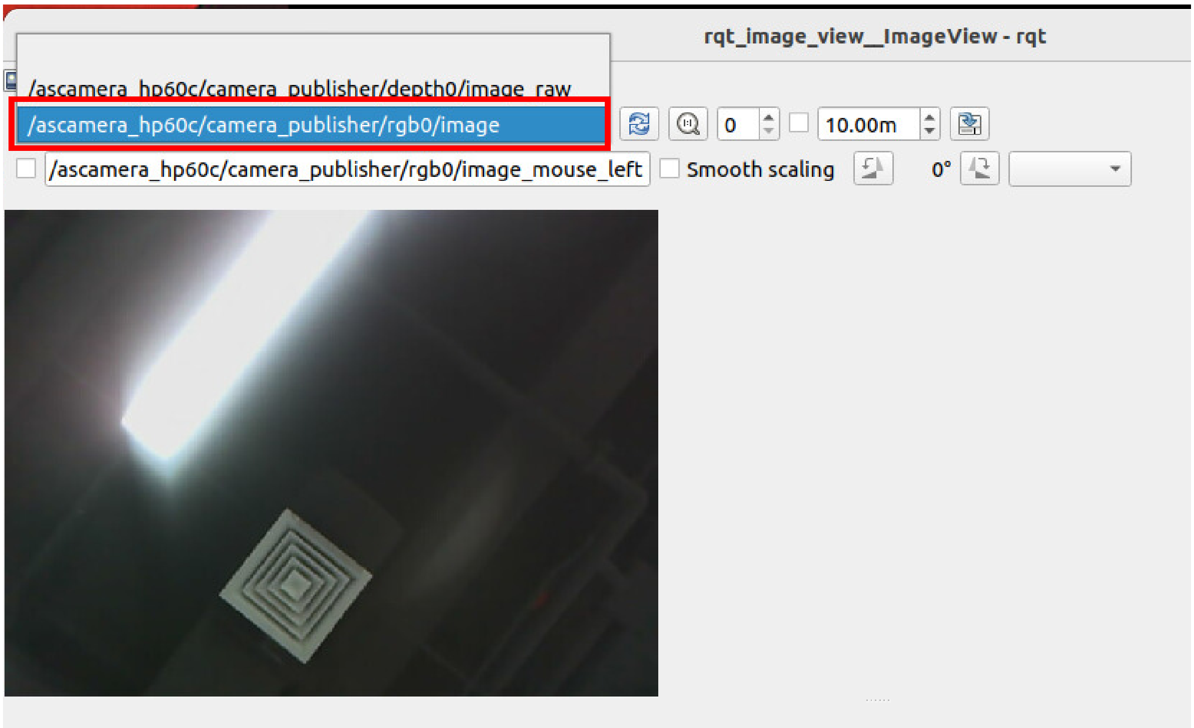
4. Subscribe to Image Topics to Display Images

4.1. Subscribe to Color Image

According to the queried node information, the color image topic can be found: /ascamera_hp60c/camera_publisher/rgb0/image Color image topic

After the camera is started, you can enter the following command in the terminal to start the rqt_images_view tool, and view images according to the selected image topic. Enter in terminal,

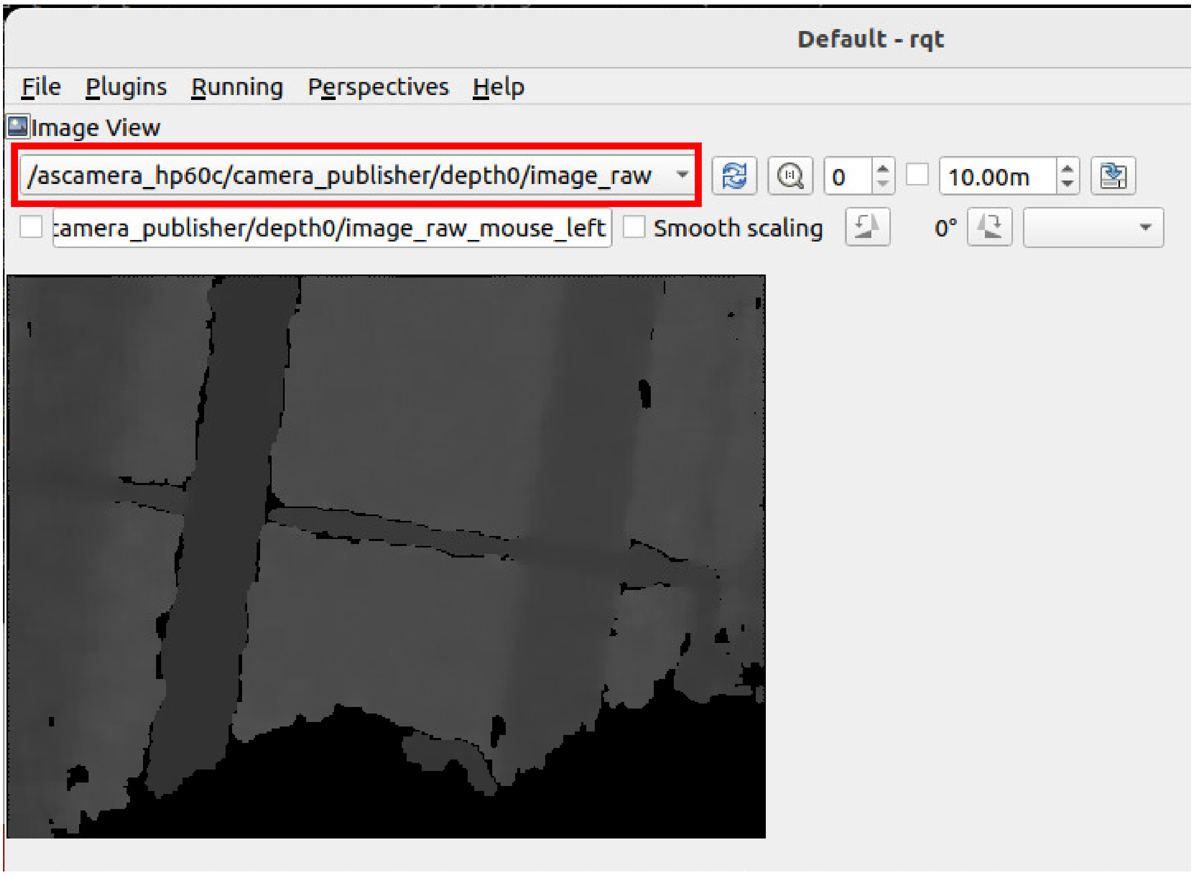
```
ros2 run rqt_image_view rqt_image_view
```



4.2. Subscribe to Depth Image

According to the queried node information, the depth image topic can be found: /ascamera_hp60c/camera_publisher/depth0/image_raw Depth image topic

Similarly, we can enter the command mentioned above `ros2 run rqt_image_view rqt_image_view` in the terminal to start the rqt_images_view tool, and select the depth image topic to display depth images,



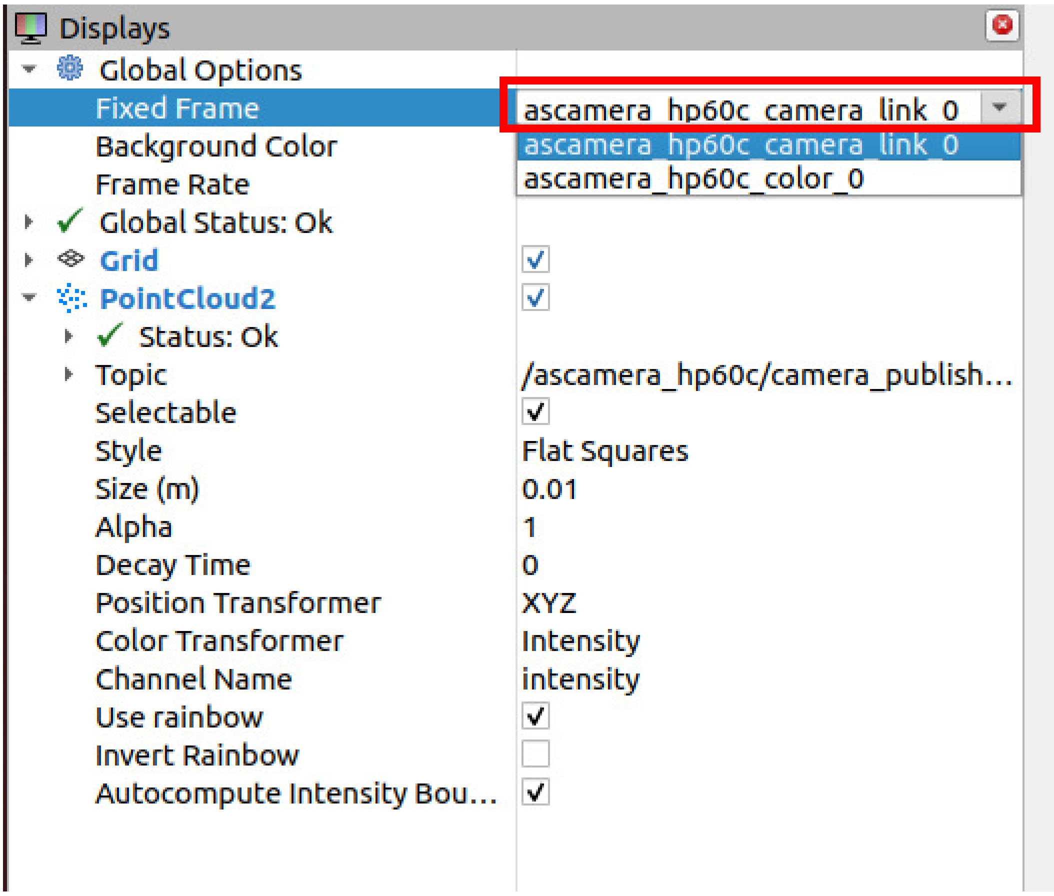
As shown in the figure above, the depth image of the /camera/depth/image_raw depth image topic is displayed.

4.3 Visualize and View Point Cloud

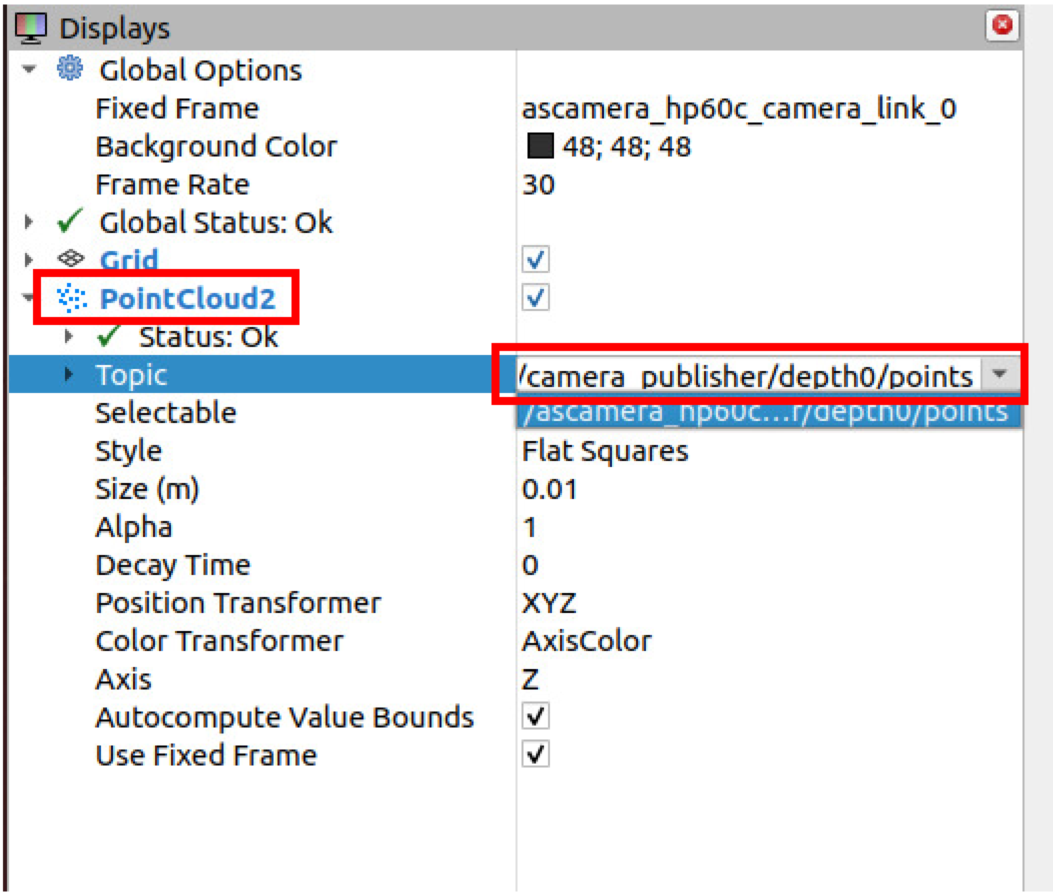
- Enter command in terminal to start RVIZ2

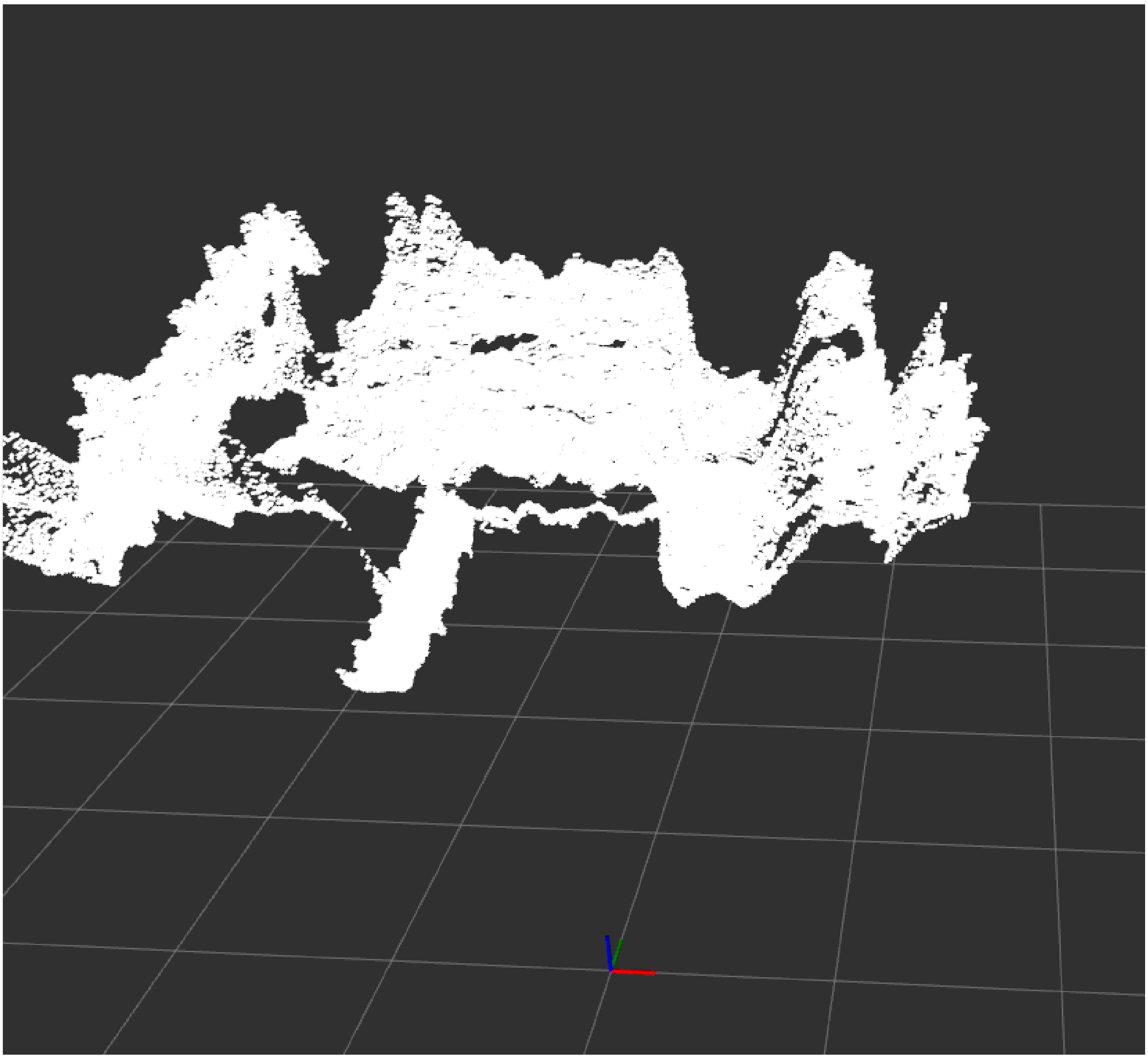
```
rviz2
```

Set the reference coordinate system to ascamera_hp60c_camera_link_0, as shown in the figure below:



- In PointCloud2, select Topic as `/ascamera_hp60c/camera_publisher/depth0/points` to view the point cloud image of the depth camera


Displays
Global Options
Fixed Frame
ascamera_hp60c_camera_link_0
Background Color
48; 48; 48
Frame Rate
30
Global Status: Ok
Grid
PointCloud2
Status: Ok
Topic
/camera_publisher/depth0/points
/ascamera_hp60c...r/depth0/points
Selectable
Style
Flat Squares
Size (m)
0.01
Alpha
1
Decay Time
0
Position Transformer
XYZ
Color Transformer
AxisColor
Axis
Z
Autocompute Value Bounds
Use Fixed Frame



- Additionally, you can modify the color of the point cloud display